

Multiple Choice

1. **Answer:** A

Options: A) Linear hard-margin SVM.; B) Linear Logistic Regression.; C) Linear Soft margin SVM.; D) The centroid method.

Note: Correct option: A - Linear hard-margin SVM.

2. **Answer:** B

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: B - False, False

3. **Answer:** A

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: A - True, True

4. **Answer:** D

Options: A) Partitioning based clustering; B) K-means clustering; C) Grid based clustering; D) All of the above

Note: Correct option: D - All of the above

5. **Answer:** C

Options: A) linear in D; B) linear in N; C) polynomial in D; D) dependent on the number of iterations

Note: Correct option: C - polynomial in D

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The use of weak classifiers'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'True, False'.

Long Form Response

11. **Answer:** 1. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** False, False. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key